Section 1 (24 points)

1) B
2) A
3) E
4) D
5) E
6) C
7) E
8) A
9) A
10) E
11) D
12) C
13) D
14) C
15) C
16) D
17) B
18) A
19) A
20) C
21) D
22) D
23) D
24) A

Section 2 (27 points)

1) Lane 2 may have supercoiled DNA (1). Supercoiled DNA occurs when the DNA helix twists on itself (1). This creates DNA molecules which are smaller than other arrangements of DNA (1); since these fragments are smaller, they are able to travel further through the agarose gel (1).

2) Dosage compensation (1) is the process by which 1 X chromosome becomes inactivated (lyonization) (1), preventing excessive amounts of X gene products from being produced (1).

3) No, these findings should not surprise them (1) because X inactivation is random (1), meaning the unhealthy X chromosome could be permanently silenced (1), resulting in no disease being present in some of the cells (1).
    a) Any mention (with an explanation) of mosaicism can earn the 3rd and 4th point.

    OR

    Yes, these findings should surprise them (1) X inactivation would not matter if the woman is carrying 2 unhealthy X chromosomes (1). X inactivation only silences 1 out of 2 X chromosomes (1) so at least 1 unhealthy X chromosome is being used to code for products (1).

4) Response **must** mention or indicate that since the frequency of alleles does not change, the population can be assumed to be under H.W. equilibrium. (1)
    a) Writing both H.W. formulas in the response earns this point

    $q=3/80=0.0375$ (1)
    $p+q=1 \rightarrow p=0.9625$ (1)
    $p^2+2pq+q^2=1$
    $(0.9625)^2+2(0.0375)(0.9625)+(0.0375)^2=1$
    Incidence of cystic fibrosis = $0.0375^2=0.00140625$ or roughly 0.0014 (1)

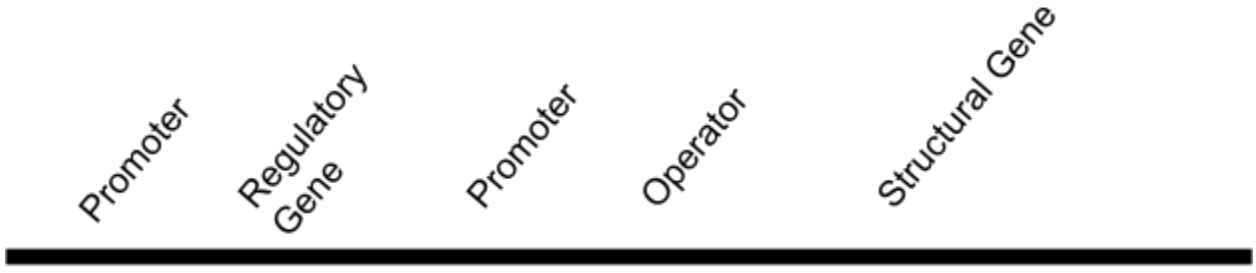
5) Acetylation increases gene expression (1). DNA is wrapped around positively charged histones. Acetylation neutralizes the positive charge (1) by adding an acetyl group to lysine residue on the histones (1). This allows for DNA to have a more relaxed chromatin structure (euchromatin) (1), making the DNA more accessible to relevant enzymes/proteins (1).

6) 1 point per correct label
    a) Leading strand
    b) Origin of replication
    c) Lagging strand
    d) RNA primer

7) AB, aB, Ab, and ab are the possible genotypes of the gamete (1). There are multiple possibilities because homologous chromosomes are independently assorted during metaphase I (1). Gametes have fewer alleles than their parents because in anaphase I, homologous chromosomes are separated and in anaphase II, sister chromatids are separated (2; 1 point for each separation mentioned).

Section 3 (45 points)

1) The central dogma of biology involves flow of information from DNA to mRNA to protein. The first step of this process is transcription. This step starts when the enzyme RNA polymerase binds to a promoter region. Two important bacterial promoters are the -10 and -35 elements. An important human promoter is the TATA box. These regions contain DNA sequences which signal the enzyme to start the process. The next step is called elongation, in which the enzyme moves in the 3' to 5' direction while adding to the mRNA strand in the 5' to 3' direction. The DNA strand that the enzyme moves down is called the template strand while the other is called the non-template/coding strand. In humans, before the next process begins, the resulting strand from the first process is altered through 3 main processes: poly-adenylation, addition of a 5' GTP cap, mRNA splicing. The final step of this process is called termination and occurs when the enzyme reaches a terminator sequence. The two main strategies for this in bacteria are rho-dependent and rho-independent. The next process in the central dogma is called translation. In bacteria, the resulting strand from the first process is read by ribosomes.
    a) Each blank is 0.5 points

2) 

    a) All-or-nothing grading; if the error is that they write the specific genes for the lac operon instead of structural gene, then deduct 1 point only. 5 points total.

3) 5' A U G C G G C A U A A G 3' → (1) for a correct mRNA sequence
    a) Indication of top strand being used for transcription (1) along with correct primes (1)
(N terminus) Formyl-methionine–arginine–histidine–lysine (C terminus)
    b) 6 total points; 1 per amino acid; 1 additional point for having the entire strand correct; 1 point for including N and C terminus in the correct places
    c) Just methionine for the first amino acid is not considered correct

4) The fluorescent gene allows us to monitor whether or not the gene is being transcribed (1) since the green promoter is only active when the operon before it is active (1). If the fluorescent protein is present, we know that the gene has been transcribed (1).

5) Standard PCR (1)
   Step 1: Denaturing involves using high temperatures to cause DNA to break into single strands (95 deg C) (1)
   Step 2: Annealing allows for primers to anneal to ssDNA at lower temperatures (55 deg C) (1)
   Step 3: Elongation involves synthesis of new strands by Taq polymerase (72 deg C) (1)
   a) Error of margin for temperatures is +/- 5 deg C
   b) Accept other PCR methods which are used for amplification of DNA (use the internet to verify temperatures with the same error margin with the same scoring rubric of 1 point per step)

6) Taq polymerase (1) is useful because it does not denature at higher temperatures than human polymerase(1), allowing for elongation to occur and creating identical copies of the DNA.
   a) If the answer to #5 is a PCR method which does not use taq polymerase specifically (e.g. pfu PCR), grade the answer as:
      i) Name of the enzyme (1)
      ii) Similar/same reasoning as the rubric (1) OR specifying an advantage of the PCR method (e.g. pfu PCR has a higher accuracy due to proofreading functions) (1)

7) Bacteria replicate faster than eukaryotic cells which allows the plasmid to be cloned many times rapidly (1). The origin of replication (ori) of plasmids are not compatible with eukaryotic cells (1).
   a) Answers must say bacterial cells to be eligible for credit.